

SAT Math Mastery · Trick Pack 1H**SAT Trick Pack 1 — Part H — Answer Key & Worked Solutions**

Time	75 minutes: 55 min solving + 20 min self-check and error log
Topics	conditional probability · line and circle tangency · cofunction identities · perpendicular lines · intercept form · translations · regular polygons · Vieta and discriminant · rational formulas · composite solids · percent ratios · altitude relations · radical equations
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Turn conditional-probability statements into linear equations in the unknown counts; the conditioned group is the denominator.
- S2.** Substitute a line into a circle or parabola and read the number of solutions from the discriminant.
- S3.** Use $\sin \theta = \cos(90^\circ - \theta)$ and the second branch $\sin A = \sin(180^\circ - A)$; filter by the allowed interval.
- S4.** Perpendicular lines in standard form: $Ax + By = C$ has slope $-\frac{A}{B}$, so $A_1A_2 + B_1B_2 = 0$.
- S5.** Intercept form with one known zero: divide the two point equations to cancel the leading coefficient.
- S6.** Read a translated graph as an evaluation of the original function at a shifted input.
- S7.** Regular polygons: base angle $\frac{180^\circ}{n}$, exterior angle $\frac{360^\circ}{n}$; match each given angle to the right triangle angle.
- S8.** Undo outside operations first, then inside, to recover points of an unknown linear function.
- S9.** Isolate a variable in a rational formula by one common denominator, or test with simple values.
- S10.** Composite solids: subtract the removed volume, convert cubic units with the cube of the length factor, multiply by density.
- S11.** Divide two percent equations that share a base to cancel it.
- S12.** Altitude to the hypotenuse: $CD^2 = AD \cdot DB$, $AC^2 = AD \cdot AB$, $BC^2 = DB \cdot AB$.
- S13.** Radical equations: square, then check every root against the sign condition on the other side.

2 · REQUIRED FACTS

- $P(A | B) = \frac{\text{count}(A \text{ and } B)}{\text{count}(B)}$: the denominator is only the group named after “given”.
- A line meets a circle or parabola in 2, 1 or 0 points as the discriminant of the substituted quadratic is positive, zero or negative.
- $\sin \theta = \cos(90^\circ - \theta)$; $\sin A = \sin B$ gives $A = B + 360^\circ n$ or $A = 180^\circ - B + 360^\circ n$.
- Two non-vertical lines are perpendicular exactly when the product of their slopes is -1 .
- $f(x - h) + k$ shifts the graph h right and k up; its y -intercept is $f(-h) + k$.
- Regular n -gon: interior angle $180^\circ - \frac{360^\circ}{n}$; an isosceles triangle on two consecutive sides has base angles $\frac{180^\circ}{n}$.
- $1 \text{ cm}^3 = 1000 \text{ mm}^3$ (the factor is cubed); mass = density $\times$ volume.
- A square-root equation $\sqrt{u} = v$ requires $v \geq 0$; squaring can create extraneous roots.
- For $x^2 + px + q$ with real roots: sum = $-p$, product = q ; the roots are non-real exactly when $p^2 - 4q < 0$.

3 · REQUIRED FORMULAS

$$\Delta = b^2 - 4ac, \quad r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}, \quad m = -\frac{A}{B}, \quad A_1A_2 + B_1B_2 = 0, \quad CD^2 = AD \cdot DB$$

$$\text{ext. angle} = \frac{360^\circ}{n}, \quad \text{base angle} = \frac{180^\circ}{n}, \quad V_{\text{cyl}} = \pi r^2 h, \quad \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}, \quad \sin \theta = \cos(90^\circ - \theta)$$

4 · MINI THEORY SHEET

Conditional counts. Write each probability as a fraction of its own group, cross-multiply, and solve the two linear equations. Never divide by the overall total unless the sentence says “of all”.

Tangency and parameters. Substitute, collect a quadratic, set $\Delta = 0$ for one solution, and only then apply any sign condition on the parameter to pick the root.

Trig equations. Convert to one function, write both branches, add multiples of 360° , and keep only the values in the stated interval.

Transformations. Outside operations (multiply, add) act on outputs, inside operations act on inputs. To recover f , undo the outside first; a point (p, q) of $g(x) = af(x - h) + k$ gives $f(p - h) = \frac{q - k}{a}$.

Checking roots. After squaring, test each root in the original equation, or impose the sign condition before counting.

Speed shortcuts.

Situation	Shortcut
“of those who ...”	the conditioned group is the denominator; cross-multiply the fractions
line tangent to a circle	$\Delta = 0$ right after substitution; then use the sign condition
$\sin(\cdot) = \cos(\cdot)$	rewrite $\cos B = \sin(90^\circ - B)$; solve both branches; filter
perpendicular, standard form	$A_1A_2 + B_1B_2 = 0$; no slopes needed
one known zero, two points	divide the two point equations to cancel a
shift right h	new y -intercept is $f(-h)$ (not $f(h)$)
regular polygon, diagonal	isosceles triangle: base $\frac{180^\circ}{n}$; apex = interior angle
transformed linear graph	undo outside first, then inside; slope from two recovered points
rational formula	plug simple values for the other letters and test the options
drilled solid and density	factor common πh , convert with the cube of the length factor
two percent equations, same base	divide: the base cancels
altitude to the hypotenuse	$\text{leg}^2 = \text{hypotenuse} \times \text{adjacent segment}$; $\text{altitude}^2 = \text{product of segments}$
$\sqrt{u} = v$ with a parameter	square, then require $v \geq 0$ for each root
quadratic factor with no real zero	strict $\Delta < 0$; test the boundary value before counting it

5 · GUIDED EXAMPLES

Example 1 — Conditional probability to a linear system

A club has 50 members, j juniors and s seniors. Half the juniors and $\frac{1}{5}$ of the seniors are in the band. A random band member is a junior with probability $\frac{5}{8}$. Find j .

$$\frac{j/2}{j/2 + s/5} = \frac{5}{8} \Rightarrow 8 \cdot \frac{j}{2} = 5 \left(\frac{j}{2} + \frac{s}{5} \right) \Rightarrow 4j = \frac{5j}{2} + s \Rightarrow s = \frac{3j}{2}.$$

$$j + s = 50 \Rightarrow \frac{5j}{2} = 50 \Rightarrow j = 20. \text{ (Check: band} = 10 + 6 = 16, \text{ juniors } \frac{10}{16} = \frac{5}{8}.)$$

Example 2 — Tangent line: discriminant first, sign second

The line $y = x + c$, $c > 0$, meets $x^2 + y^2 = 8$ in exactly one point. Find c .

$$x^2 + (x + c)^2 = 8 \Rightarrow 2x^2 + 2cx + c^2 - 8 = 0. \Delta = 4c^2 - 8(c^2 - 8) = 64 - 4c^2 = 0 \Rightarrow c^2 = 16.$$

$c = \pm 4$; the condition $c > 0$ gives $c = 4$.

Example 3 — Radical equation: square, then test the sign

Solve $\sqrt{x + 6} = x$.

$$\text{Square: } x + 6 = x^2 \Rightarrow x^2 - x - 6 = 0 \Rightarrow x = 3 \text{ or } x = -2.$$

The right side must be ≥ 0 , so $x = -2$ is extraneous ($\sqrt{4} = 2 \neq -2$). Only $x = 3$ works.

6 · PRACTICE SET — 16 PROBLEMS · ALL SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · NO CALCULATOR NEEDED
A survey of 240 adults asked whether they commute by bicycle and whether they own a helmet. Of the adults who commute by bicycle, $\frac{3}{5}$ own a helmet. Of the adults who do not commute by bicycle, $\frac{1}{4}$ own a helmet. If one adult is selected at random from those who own a helmet, the probability that the selected adult commutes by bicycle is $\frac{12}{19}$. How many of the 240 adults own a helmet?

- (A) 60 (B) 95 ✓
(C) 144 (D) 100

Worked solution

- Let c be the number of bicycle commuters, so $240 - c$ do not commute by bicycle. Helmet owners: $\frac{3}{5}c$ commuters and $\frac{1}{4}(240 - c)$ non-commuters.
- The condition “given a helmet owner” makes all helmet owners the denominator: $\frac{\frac{3}{5}c}{\frac{3}{5}c + \frac{1}{4}(240 - c)} = \frac{12}{19}$.
- Cross-multiply: $19 \cdot \frac{3}{5}c = 12 \left(\frac{3}{5}c + \frac{1}{4}(240 - c) \right) \Rightarrow \frac{57}{5}c - \frac{36}{5}c = 720 - 3c$, so $\frac{21}{5}c + 3c = 720$ and $\frac{36}{5}c = 720$, giving $c = 100$.
- Helmet owners: $\frac{3}{5}(100) + \frac{1}{4}(140) = 60 + 35 = 95$. Check: $\frac{60}{95} = \frac{12}{19}$.
- The number of helmet owners is 95. (The options 60, 144 and 100 are the commuters with helmets, $\frac{3}{5}$ of 240, and the number of commuters.)

Answer: (B)

2. SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 = 45$$

$$y = mx + 15$$

In the system above, m is a negative constant. If the system has exactly one solution (x, y) , what is the value of x ?

- (A) -3 (B) 3
(C) 6 ✓ (D) -6

Worked solution

- Substitute: $x^2 + (mx + 15)^2 = 45 \Rightarrow (1 + m^2)x^2 + 30mx + 180 = 0$.
- Exactly one solution means $\Delta = 0$: $900m^2 - 4(1 + m^2)(180) = 0 \Rightarrow 900m^2 - 720 - 720m^2 = 0 \Rightarrow 180m^2 = 720 \Rightarrow m^2 = 4$.
- The condition $m < 0$ gives $m = -2$ (not $m = 2$).
- The quadratic becomes $5x^2 - 60x + 180 = 0$, i.e. $x^2 - 12x + 36 = 0$, so $(x - 6)^2 = 0$ and $x = 6$.
- Check: $y = -2(6) + 15 = 3$ and $6^2 + 3^2 = 45$. The value of x is 6 (the choice $m = 2$ would give $x = -6$).

Answer: (C)

3.

SAT CHALLENGE · CALCULATOR

For $0 < x < 45$, the equation

$$\sin((5x - 20)^\circ) = \cos((x + 14)^\circ)$$

has more than one solution. What is the sum of all solutions x in this interval?**Worked solution**

1. Write the cosine as a sine: $\cos((x + 14)^\circ) = \sin((76 - x)^\circ)$.
2. Equal sines give two branches. Branch 1: $5x - 20 = 76 - x + 360n \Rightarrow 6x = 96 + 360n$. Only $n = 0$ lies in $(0, 45)$: $x = 16$.
3. Branch 2: $5x - 20 = 180 - (76 - x) + 360n = 104 + x + 360n \Rightarrow 4x = 124 + 360n$. Only $n = 0$ lies in $(0, 45)$: $x = 31$.
4. Check: $x = 16$ gives $\sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$; $x = 31$ gives $\sin 135^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}$. Both satisfy the equation.
5. The sum is $16 + 31 = 47$.

Answer: 47

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, line ℓ has equation $5x - 2y = 11$ and line m has equation $ax + by = d$, where a , b and d are positive constants and $a + b = 21$. Lines ℓ and m are perpendicular, and line m passes through the point $(4, 3)$. What is the value of d ?

Worked solution

1. Perpendicular lines in standard form satisfy $A_1A_2 + B_1B_2 = 0$: $5a + (-2)b = 0$, so $5a = 2b$ and $b = \frac{5}{2}a$.
2. (Slope check: ℓ has slope $\frac{5}{2}$, m has slope $-\frac{a}{b} = -\frac{2}{5}$, and the product is -1 .)
3. $a + b = 21 \Rightarrow a + \frac{5}{2}a = 21 \Rightarrow \frac{7}{2}a = 21 \Rightarrow a = 6$, so $b = 15$.
4. Since $(4, 3)$ lies on m : $d = 6(4) + 15(3) = 24 + 45 = 69$.

Answer: 69

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has a zero at $x = 2$. The graph of $y = f(x)$ in the xy -plane passes through the points $(0, -16)$ and $(5, 54)$. What is the minimum value of f ?

(A) -18 ✓(B) -16 (C) -9 (D) -1 **Worked solution**

1. Let the other zero be s and write $f(x) = a(x - 2)(x - s)$.
2. $f(0) = a(-2)(-s) = 2as = -16$ and $f(5) = a(3)(5 - s) = 54$.
3. Divide the second equation by the first to cancel a : $\frac{3(5 - s)}{2s} = \frac{54}{-16} = -\frac{27}{8}$. Then $24(5 - s) = -54s$, so $120 = -30s$ and $s = -4$.
4. Then $2a(-4) = -16 \Rightarrow a = 2 > 0$, so the parabola opens upward and has a minimum.
5. The vertex is at $x = \frac{2 + (-4)}{2} = -1$: $f(-1) = 2(-3)(3) = -18$.
6. The minimum value is -18 . (The options -16 and -1 are $f(0)$ and the vertex's x -coordinate.)

Answer: (A)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of the quadratic function f in the xy -plane passes through the points $(0, 1)$, $(1, -2)$ and $(3, 4)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ 3 units to the right and 5 units down. What is the y -coordinate of the y -intercept of the graph of g ?

(A) 34

(B) 39

(C) -1

(D) 29 ✓

Worked solution

1. Let $f(x) = ax^2 + bx + c$. From $(0, 1)$: $c = 1$.

2. From $(1, -2)$: $a + b + 1 = -2 \Rightarrow a + b = -3$. From $(3, 4)$: $9a + 3b + 1 = 4 \Rightarrow 3a + b = 1$.

3. Subtract: $2a = 4$, so $a = 2$ and $b = -5$. Thus $f(x) = 2x^2 - 5x + 1$.

4. The translation gives $g(x) = f(x - 3) - 5$, so the y -intercept is $g(0) = f(-3) - 5$.

5. $f(-3) = 2(9) + 15 + 1 = 34$, so $g(0) = 34 - 5 = 29$.

6. The answer is 29. (The options 34 and 39 forget the shift down or reverse it; -1 uses $f(3) - 5$, the wrong direction of the horizontal shift.)

Answer: (D)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

Points A , B , C and D are consecutive vertices of a regular polygon with n sides, and $\overline{AC}$ is a diagonal. The measure of $\angle BAC$ is $(x + 3)^\circ$ and the measure of $\angle ACD$ is $(10x + 15)^\circ$, where x is a constant. What is the value of n ?

(A) 24

(B) 8

(C) 12 ✓

(D) 15

Worked solution

1. Triangle ABC is isosceles with $AB = BC$ and apex angle $\angle ABC$ equal to the interior angle $180^\circ - \frac{360^\circ}{n}$.
So $\angle BAC = \angle BCA = \frac{180^\circ}{n}$.

2. The interior angle at C is $\angle BCD = 180^\circ - \frac{360^\circ}{n}$, and it is split by the diagonal: $\angle ACD = \angle BCD - \angle BCA = 180^\circ - \frac{540^\circ}{n}$.

3. Equate with the given expressions: $x + 3 = \frac{180}{n}$ and $10x + 15 = 180 - \frac{540}{n}$.

4. Substitute $x = \frac{180}{n} - 3$: $\frac{1800}{n} - 30 + 15 = 180 - \frac{540}{n} \Rightarrow \frac{2340}{n} = 195 \Rightarrow n = 12$.

5. Check: $\angle BAC = 15^\circ$ gives $x = 12$; $\angle ACD = 10(12) + 15 = 135^\circ = 180^\circ - \frac{540^\circ}{12}$.

6. $n = 12$. (Treating 135° as the interior angle gives 8; treating 15° as the exterior angle gives 24.)

Answer: (C)

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is linear. The function g is defined by $g(x) = -2f(x - 1) + 5$, and the graph of $y = g(x)$ passes through the points $(0, 3)$ and $(2, -5)$. The graph of $y = h(x)$ is the graph of $y = f(x)$ translated 2 units to the left and 6 units up. What is the y -coordinate of the y -intercept of the graph of h ?

Worked solution

1. Undo the outside operations first. From $g(0) = 3$: $-2f(-1) + 5 = 3 \Rightarrow f(-1) = 1$.
2. From $g(2) = -5$: $-2f(1) + 5 = -5 \Rightarrow f(1) = 5$. (The inputs of f are $x - 1$, not x .)
3. Slope of f : $\frac{5 - 1}{1 - (-1)} = 2$. Then $f(x) = 2x + b$ and $5 = 2(1) + b$ gives $b = 3$: $f(x) = 2x + 3$.
4. Translating left 2 and up 6 gives $h(x) = f(x + 2) + 6$.
5. $h(0) = f(2) + 6 = 7 + 6 = 13$.

Answer: 13

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$y = 3x^2 + 4$$

$$y = 1 - kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x, y) whose x -coordinates are both negative?

(A) 6

(B) 7 ✓(C) -7

(D) 2

Worked solution

1. Set the expressions equal: $3x^2 + 4 = 1 - kx \Rightarrow 3x^2 + kx + 3 = 0$.
2. Two distinct real solutions: $\Delta = k^2 - 36 > 0$, so $|k| > 6$. This excludes $k = 2$ and the tangent case $k = 6$.
3. Both roots negative: the sum $-\frac{k}{3} < 0$ requires $k > 0$. (The product $\frac{3}{3} = 1 > 0$ already makes the roots the same sign.)
4. Together: $k > 6$. Of the options only $k = 7$ qualifies; $k = -7$ gives two distinct solutions that are both positive.

Answer: (B)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{2}{a} - \frac{3}{b} = \frac{5}{c}$$

In the equation above, a , b and c are nonzero constants and $2c \neq 5a$. Which of the following expresses b in terms of a and c ?

(A) $b = \frac{3ac}{5a - 2c}$

(B) $b = \frac{2c - 5a}{3ac}$

(C) $b = \frac{3ac}{2c + 5a}$

(D) $b = \frac{3ac}{2c - 5a}$ ✓

Worked solution

1. Isolate the b -term: $\frac{3}{b} = \frac{2}{a} - \frac{5}{c}$.

2. Common denominator on the right: $\frac{2}{a} - \frac{5}{c} = \frac{2c - 5a}{ac}$.

3. So $\frac{3}{b} = \frac{2c - 5a}{ac}$. Take reciprocals and multiply by 3: $b = \frac{3ac}{2c - 5a}$.

4. Check with $a = 1$, $c = 5$: $2 - \frac{3}{b} = 1 \Rightarrow b = 3$, and $\frac{3(5)}{10 - 5} = 3$. The reversed-sign option gives -3 , the inverted option gives $\frac{1}{3}$, and $\frac{15}{15} = 1$ for the last.

Answer: (D)

11.

SAT CHALLENGE · CALCULATOR

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm = 10 mm)

(A) 821 ✓

(B) 882

(C) 82,090

(D) 784

Worked solution

1. Volume of the full cylinder: $\pi(30)^2(40) = 36,000\pi \text{ mm}^3$.

2. Volume of the hole (radius 10, depth 25): $\pi(10)^2(25) = 2,500\pi \text{ mm}^3$.

3. Remaining volume: $36,000\pi - 2,500\pi = 33,500\pi \text{ mm}^3$.

4. Convert with the cube of the length factor: $1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$, so $V = 33.5\pi \approx 105.2 \text{ cm}^3$.

5. Mass = $7.8 \times 33.5\pi = 261.3\pi \approx 820.9 \approx 821$ grams.

6. (Other options: 784 drills the hole through the full height, 882 ignores the hole, 82,090 divides by 10 instead of 1000.)

Answer: (A)

12.

SAT CHALLENGE · CALCULATOR

A retailer's total sales were the same in March and April. In March, online sales were p percent of the total sales and came to \$18,000. In April, online sales were $(p + 8)$ percent of the total sales and came to \$25,200. What were the retailer's total sales, in dollars, for one of these months?

Worked solution

1. Let T be the total sales. March: $\frac{p}{100}T = 18000$. April: $\frac{p+8}{100}T = 25200$.
 2. Divide April by March; T cancels: $\frac{p+8}{p} = \frac{25200}{18000} = \frac{7}{5}$.
 3. Cross-multiply: $5p + 40 = 7p \Rightarrow p = 20$.
 4. Then $\frac{20}{100}T = 18000 \Rightarrow T = 90,000$. Check: 28% of 90,000 is 25,200.
- Answer:** 90000

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The length of $\overline{CD}$ is 6, and DB is 5 units longer than AD . What is the value of $\sin B$?

(A) $\frac{6}{13}$

(B) $\frac{2\sqrt{13}}{13}$ ✓

(C) $\frac{3\sqrt{13}}{13}$

(D) $\frac{4}{13}$

Worked solution

1. Let $AD = t$, so $DB = t + 5$. The altitude to the hypotenuse satisfies $CD^2 = AD \cdot DB$: $36 = t(t + 5)$.
2. $t^2 + 5t - 36 = 0 \Rightarrow (t + 9)(t - 4) = 0$, so $t = 4$ (lengths are positive). Then $AD = 4$, $DB = 9$, $AB = 13$.
3. Leg relation: $BC^2 = DB \cdot AB = 9 \cdot 13 = 117$, so $BC = 3\sqrt{13}$.
4. $\sin B = \frac{AC}{AB}$ or, in triangle CDB , $\frac{CD}{CB} = \frac{6}{3\sqrt{13}} = \frac{2}{\sqrt{13}} = \frac{2\sqrt{13}}{13}$.
5. (Check: $AC^2 = AD \cdot AB = 52$, $AC = 2\sqrt{13}$, and $\frac{2\sqrt{13}}{13}$ again. The option $\frac{3\sqrt{13}}{13}$ is $\sin A$, $\frac{4}{13}$ is $\cos^2 A$ and $\frac{6}{13}$ is $\frac{CD}{AB}$.)

Answer: (B)

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is linear. The function g is defined by $g(x) = \frac{f(x+2) - x}{x-3}$ for $x \neq 3$. Some values of g are shown in the table below.

x	1	5
$g(x)$	3	-11

What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

(A) -5

(B) -4

(C) 4 ✓

(D) -3

Worked solution

1. At $x = 1$: $3 = \frac{f(3) - 1}{1 - 3} = \frac{f(3) - 1}{-2} \Rightarrow f(3) - 1 = -6 \Rightarrow f(3) = -5$.

2. At $x = 5$: $-11 = \frac{f(7) - 5}{5 - 3} = \frac{f(7) - 5}{2} \Rightarrow f(7) - 5 = -22 \Rightarrow f(7) = -17$.

3. Slope of f : $\frac{-17 - (-5)}{7 - 3} = \frac{-12}{4} = -3$.

4. $f(3) = -3(3) + b = -5 \Rightarrow b = 4$. So $f(x) = -3x + 4$ and the y -intercept is 4.

5. (The option -3 is the slope, -5 is $f(3)$.)

Answer: (C)

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

For how many integer values of k with $-10 \leq k \leq 10$ does the equation

$$\sqrt{x+k} = x-2$$

have exactly one real solution?

Worked solution

1. Square both sides, remembering the condition $x - 2 \geq 0$: $x + k = x^2 - 4x + 4$, so $x^2 - 5x + (4 - k) = 0$ with roots $x = \frac{5 \pm \sqrt{9 + 4k}}{2}$.

2. Real roots need $9 + 4k \geq 0$; for integers this means $k \geq -2$. For $k \leq -3$ there are no solutions.

3. The larger root is at least $\frac{5}{2} > 2$, so it is always valid. The smaller root is valid only if $\frac{5 - \sqrt{9 + 4k}}{2} \geq 2$, i.e. $\sqrt{9 + 4k} \leq 1$, i.e. $k \leq -2$.

4. $k = -2$: the roots are 2 and 3, and both work, so there are two solutions. For $k \geq -1$ the smaller root is below 2 and is extraneous, leaving exactly one solution.

5. The values are $k = -1, 0, 1, \dots, 10$, which is 12 integers.

Answer: 12

16.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 + bx^2 + cx + 12$ has integer coefficients b and c , and $x + 3$ is a factor of $p(x)$. If p has exactly one real zero, what is the greatest possible value of c ?

Worked solution

1. $x + 3$ is a factor, so $p(-3) = 0$: $-27 + 9b - 3c + 12 = 0 \Rightarrow c = 3b - 5$.
2. Divide: $p(x) = (x + 3)(x^2 + ux + 4)$ with $u = b - 3$. (Expanding gives $x^3 + (u + 3)x^2 + (3u + 4)x + 12$, so $b = u + 3$ and $c = 3u + 4 = 3b - 5$.)
3. Exactly one real zero means the quadratic factor has no real zero: $u^2 - 16 < 0$, a strict inequality, so $-4 < u < 4$.
4. $u = 4$ is excluded: $x^2 + 4x + 4 = (x + 2)^2$ has the real zero -2 , giving two distinct real zeros of p (-3 and -2).
5. The greatest integer u is 3, so $b = 6$ and $c = 3(6) - 5 = 13$. (Including the boundary $u = 4$ would wrongly give $c = 16$.)

Answer: 13